

MAM2000W 2IA Introductory Algebra

Syllabus achieved

The syllabus for the Final Exam includes Chapter 1, 2, 3, 4, 5 (up and including Section 5.2) from the 2IA Course notes. Although you need to study everything, here I will emphasise the most important definitions and results for the final exam

In the exam, you will be asked to complete a proof from the list below. Make sure you understand and study these. You might be asked to state a few of the statements that you will find below. Please make sure you learn all of these!

1. Definitions

- Gcd. Prime and coprime integers.
 - Residue classes modulo n . Integers modulo n .
 - Permutations. Identity permutations.
 - Matrix-notation of a permutation.
 - Disjoint permutations and examples.
 - Cycles (and transpositions).
 - Even (respectively, odd) permutations and examples.
 - Groups. Examples of abelian groups; examples of non-abelian groups.
 - Order of an element in a group.
 - Subgroups and examples.
 - Cyclic groups and examples.
 - Cyclic subgroup generated by an element.
 - Group homomorphisms, epimorphisms, monomorphisms, isomorphisms, automorphisms.
 - Left and right cosets.
 - Index of a subgroup in a group.
 - Normal subgroups. Examples of normal subgroups.
 - Factor groups.
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2. Statements, algorithms and more

- Bézout's Lemma (very important!). Recall that the converse is not true if $d > 1$.
 - Prime Factorisation Theorem.
 - Modular arithmetic.
 - Solving equations in $\mathbb{Z}_n$.
 - Classical Chinese Remainder Theorem.
 - Matrix-type notation for permutations. Cycle notation.
 - Product of permutations.
 - Inverse of a permutation.
 - How to factor a permutation into disjoint cycles.
 - How to factor a cycle into a product of transpositions.
 - Cycle Decomposition Theorem.
 - Parity Theorem.
 - Inverses, exponent and cancellation laws.
 - Subgroup test.
 - Description of the cyclic subgroup generated by an element (Theorem 3.25).
 - Properties of cyclic groups (Theorem 3.28).
 - Fundamental Theorem of Finite Cyclic groups (definitely, very important!)
 - Describing the subgroups of a finite group, and drawing the corresponding lattice diagram (super important!)
 - Properties of group homomorphisms (Theorem 3.38).
 - Isomorphism of cyclic groups (Theorem 3.42).
 - Cosets properties (very important!).
 - Lagrange's Theorem (very important!).
 - Normality Test.
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3. Proofs to complete

You will be asked to complete one of the following proofs:

- Chinese Remainder Theorem.
- Fundamental Theorem of Finite Cyclic Groups.
- Lagrange's Theorem.